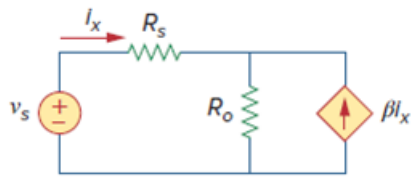


Problem

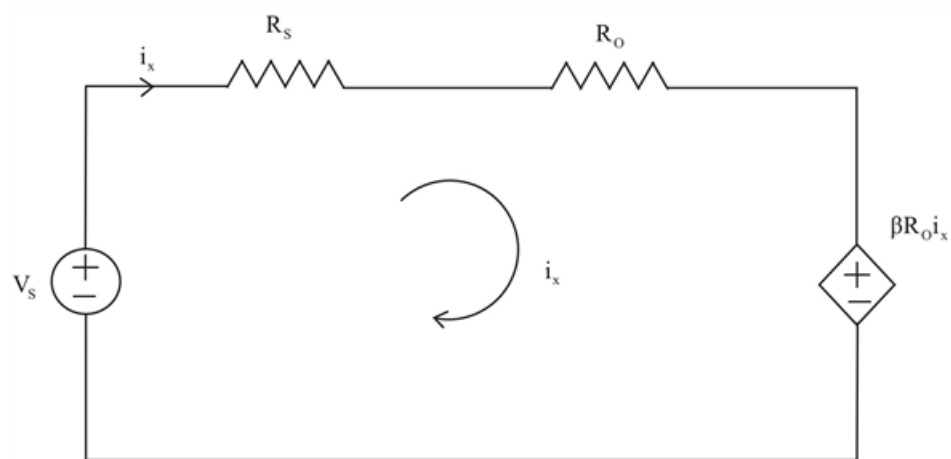
The circuit in Fig. 4.149 models a common-emitter transistor amplifier. Find i_x using source transformation.

Figure. 4.149



Step-by-step solution

Step 1 of 2



Step 2 of 2

$$-V_s + (R_s + R_o)i_x + \beta R_o i_x = 0 .$$

$$i_x = \frac{V_s}{(R_s + (1 + \beta) R_o)} .$$